

Dynamic Graph Attention: Unraveling Spatio-Temporal Synchrony in EEG Data

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Abstract—In this paper, we propose a deep model based on graph convolutional networks for emotion recognition using EEG data. The model encodes spatial and temporal features of EEG channels and learns relationships between nodes through a self-attention mechanism, capturing spatio-temporal synchrony in brain regions. Experimental results show that our model outperforms existing approaches, with the attention mechanism contributing significantly to classification accuracy. In particular, the attention scores provide insights into how EEG channels influence each other at different times, revealing spatio-temporal patterns of brain connectivity related to emotions.

Index Terms—EEG, Graph neural network, Self-Attention

I. INTRODUCTION

The analysis of electroencephalography (EEG) data holds immense significance in understanding the underlying neural processes and deciphering human cognitive and emotional states. EEG, indeed, provides a unique window into the functioning of the brain, offering real-time measurements of electrical activities generated by neurons. Its non-invasive nature and high temporal resolution make it an invaluable tool for investigating brain dynamics during various cognitive tasks and emotional experiences. However, the spatio-temporal complexity of EEG data, which encompasses low spatial resolution and low signal-to-noise ratio, makes it particularly challenging to analyze and interpret.

In this context, we present a novel deep model based on graph convolutional networks that addresses the challenges of interpreting electroencephalography (EEG) data for emotion recognition. Our proposed model leverages the representational power of graph convolutional networks to effectively capture the intricate relationships between EEG channels, which encode features both in spatial and temporal dimensions. Each EEG channel is encoded as a node in the graph, and our model learns to extract and exploit the intrinsic dependencies and interactions between these nodes. This process enables the network to discern patterns of neural activity that are critical for emotion recognition. To effectively capture the dynamic synchronization of brain regions involved in emotional processing, we design a specific self-attention mechanism that allows the model to selectively focus on crucial regions and time points, enhancing the interpretation of brain synchrony processes. By doing so, our model can better capture the underlying spatio-temporal neural patterns associated with different emotional states.

The results of our experiments on emotion recognition tasks demonstrate the superior performance of our proposed model compared to existing approaches.

Furthermore, to gain a deeper understanding of the model's inner workings and to identify the most influential components, we conduct ablation studies that indicate the role of the proposed attention mechanism in driving classification accuracy. This highlights the significance of considering brain synchrony processes when attempting to recognize emotions from EEG.

In summary, our proposed deep model based on graph convolutional networks, coupled with a self-attention mechanism, presents a significant advancement in the field of emotion recognition from EEG data. By effectively capturing and interpreting spatio-temporal patterns of neural activity, our model not only outperforms existing approaches but it may also shed light on the crucial role of brain synchrony processes in understanding emotional states.

II. RELATED WORK

In the last decades, EEG data analysis has been primarily employed for brain computer interface (BCI) applications (e.g., motor imagery, speech imagery, emotion recognition [1], [2]), for clinical applications (e.g., epilepsy detection and prediction [3], [4]), or for monitoring cognitive functions (mental workload, engagement, fatigue [5], [6]). Before the advent of deep learning, EEG classification was mainly accomplished by extracting ad-hoc spectral or time features (e.g., power spectral density and wavelet) and feeding them into shallow learning models such as support vector machines. One of the first deep learning approaches for learning EEG data is the EEGNet model [7], which employs depth-wise separable 1D and 2D convolutions and has become a common deep baseline for EEG classification. Despite its simplicity, EEGNet has proven effective in a variety of tasks, including P300 detection, visual-evoked potentials, error-related negativity responses, movement-related cortical potentials, and sensory motor rhythms. However, EEGNet suffers from a major limitation, as it ignores spectral and time coherence among brain regions. Spectral coherence is a popular feature in EEG data analysis and consists of pairwise channel comparisons of the amplitude and phase of the spectral data that can be used to discover functional connectivity specific to the task via synchrony patterns [8]. However, synchrony needs to be considered when dealing

with higher-order cognitive processes such as those involved in emotions, whose underlying neural mechanisms may span multiple cortical regions and are distributed over time. SyncNet [9] explicitly addresses synchrony in EEG data analysis through an approach based on biologically-inspired convolution kernels. Nevertheless, we argue that targeting only spectral coherence is insufficient, since the time dimension must also be taken into account to support modeling of long-range temporal correlations. ChannelNet [10] addresses the time dimension by aggregating features at different time scales via convolutions with different dilation values. Our work integrates the capabilities of these approaches and goes beyond them, by introducing spatio-temporal attention for adaptive modeling of long-range relationships in the data.

Graph neural networks (GNNs) have been used to learn functional connectivity (synchrony) for emotion recognition in EEG data: [11] have recently used a GNN with a sparse adjacency matrix to capture both local and global inter-channel relations based on the biological topology of the human brain. This is in contrast with CNNs [1], [12], [13] and GNNs [14] that learn inter-channel correlations using operators with narrow local receptive fields, thus excluding long-range dependencies. However, most GNN-based methods take into account relations across the spatial dimension only, while it would also be useful to model temporal relationships as many higher-order cognitive functions are typically exerted via frontal-to-posterior synchrony: for instance, visual scene processing first activates the visual cortex (area V1), but then propagates to the inferotemporal cortex (IT) many tens to hundreds of milliseconds later. We therefore extend the approach of [11] by modeling long-range spatial dependencies as well as long-range spatio-temporal correlations, and by defining channel connectivity through two graphs with different adjacency matrices, one modeling brain topology and distance among channels over the scalp, and another modeling process-related temporal correlation through self-attention. Attention mechanisms have been already used in combination to graph neural networks (GAT [15]), but these limit the attention span to a small subset of spatially-close nodes (first-order neighbors), ignoring correlations over time. Learning on dynamic graphs is relatively recent and few recent methods address the problem of continuous-time scenario [16]. In this case, the assumption is that the graph structure is dynamic, i.e., nodes and their edges can vary with time, but their adjacency in the graph lies on topological constraints. This is different from our case where graph structure — in terms of nodes — is invariant (as it depends on the brain electrodes) and edges and their adjacency are based on a spatio-temporal attention mechanism. The aspect of temporal attention in continuous-time dynamic graphs is presented in [17], but it is again limited to a subset of neighbor nodes. Our approach advances these techniques by modeling long-range spatio-temporal relationships among all graph nodes and for all time steps. Furthermore, our approach is inherently interpretable: exploring the attention scores allows neuroscientists to study the interactions between brain regions and the way these connect in time.

III. DYNAMIC GRAPH ATTENTION (DGA) MODEL

A. Overview

The Dynamic Graph Attention (DGA) architecture is presented in Fig. 1. The input EEG signal, after appropriate pre-processing, is fed to a Gaussian Process (GP) adapter [9] that maps the input channels to a smaller number of new channels. This reduces redundancy in the signal from nearby electrodes. The output of the GP adapter is a set of interpolated raw EEG signals. Temporal features are then extracted through separable 1D convolutions, to allow for learning activity patterns that are specific to each channel. Then, a non-local block [18] processes the extracted features to estimate relationships between features in different channels and at different times. This yields a new set of features and a matrix of corresponding attention scores.

Next, the model forks into two graph convolutional branches, with different connectivity patterns. In the first branch, node features are initialized to self-attended spatio-temporal features and the adjacency matrix is based on 3D electrode distances. This branch is designed to capture spatial locality, while employing features that non-locally depend on other space-time locations. In the second branch, node features are initialized to the temporal features computed through 1D separable convolutions (thus, they are channel-specific), and are non-locally combined by employing an adjacency matrix that is based on the self-attention scores provided by the non-local block. The two paths therefore independently extract features and graph edges, with both branches being able to capture global correlations, while making sure that local and long-distance contributions are also taken into account. Finally, node output features from both graphs are summed and processed by a cascade of fully-connected layers to obtain the predicted classification output.

In the following subsections, we describe each of the modules in detail.

B. Gaussian Process (GP) adapter

Since EEG cap montages (i.e., channel layouts) may vary from lab to lab as well as signal quality may vary from recording to recording (excessively high electrode impedances will reduce signal quality, especially in warmer environments: see [19]), different sessions are likely to have different subsets of valid EEG channels. Moreover, nearby electrodes are highly correlated and contain redundant information, increasing the dimensionality without providing additional classifiable data.

We therefore pass the input data through a GP adapter in the spirit of [9] and [20]. A GP adapter is a multi-task regressor, where “tasks” are defined by the estimation of channel samples at different times, mapping an EEG signal $\mathbf{X} \in \mathbb{R}^{C \times T}$, with T samples for each of C channels at locations $\mathbf{p}$, to the pseudo-signal $\mathbf{X}^* \in \mathbb{R}^{L \times T}$, where $L < C$ is the number of pseudo-inputs regressed to locations $\mathbf{p}^*$.

Assuming that each GP has zero mean (as can be expected after channel-based standardization) and that only spatial correlations are modeled, when evaluated at a finite set of

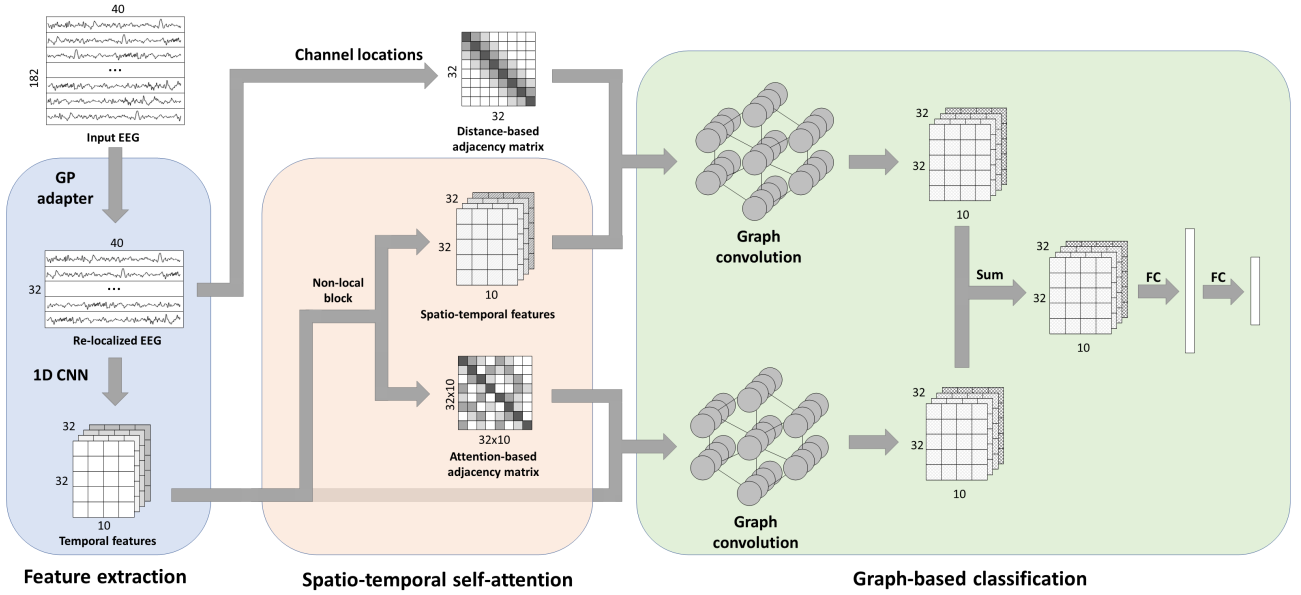


Fig. 1: Overview of the DGA architecture. The input EEG signal is fed to a Gaussian Process adapter, which reduces the dimensionality of the input. Temporal features are extracted through separable 1D convolutions, to allow for learning activity patterns that are specific for each channel. A non-local block processes the extracted features to estimate spatio-temporal features and produce a matrix of attention weights that relate feature vectors across different channels and times. Two branches of graph convolutional layers are then applied. See text for details.

points the multi-task GP can be written as a multivariate normal distribution [21]:

$$\text{vec}(\mathbf{X}) \sim \mathcal{N}(\mathbf{f}, \sigma^2 \mathbf{I}_{C \times T}), \quad \mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_{\mathbf{pp}} \otimes \mathbf{I}_T) \quad (1)$$

where $\text{vec}(\mathbf{X})$ is the vectorized data matrix $\mathbf{X}$, σ^2 is an arbitrary noise variance, and $\mathbf{K}_{\mathbf{pp}} \in \mathbb{R}^{C \times C}$ models spatial relationships across channels using the exponential kernel:

$$K_{\mathbf{p}_c \mathbf{p}_{c'}} = \alpha_1 \exp(-\alpha_2 d(\mathbf{p}_c, \mathbf{p}_{c'})) \quad (2)$$

where d is a distance metric between electrode locations (discussed in the following).

Let the pseudo-input locations be defined as $\mathbf{p}_l^*$ for $l = 1, \dots, L$. Using the GP formulation, the signal can be inferred at the L pseudo-input locations from the original signal. Following [20], the variance is ignored, and only the expectation of the signal is used (to speed up computation), which is given by:

$$\mathbf{X}^* = \mathbb{E}(\mathbf{X}^* | \mathbf{X}) = \mathbf{K}_{\mathbf{p}^* \mathbf{p}} (\mathbf{K}_{\mathbf{pp}} + \sigma^2 \mathbf{I}_C)^{-1} \mathbf{X} \quad (3)$$

Note that the whole process is differentiable with respect to the pseudo-input locations, thus allowing the model to learn the optimal locations of the new regressed channels with standard backpropagation [22].

Distances for computing $\mathbf{K}_{\mathbf{pp}}$ and $\mathbf{K}_{\mathbf{p}^* \mathbf{p}}$ are constructed for each electrode pair. Since EEG electrodes are distributed over the scalp in a three-dimensional space, we computed the electrode distances (both the original and the pseudo-inputs) via azimuthal equidistant projection (AEP). In AEP, 3D distances over a spherical surface from the center of a projection to any

other point are preserved as linear distances on the projection plane. In our case, we assume that the shape of the cap worn on a subject's head can be approximated by a sphere and the same method can be used to compute the projection of electrode locations on a 2D surface for measuring distances.

C. Temporal feature extraction

The re-localized EEG, provided as output by the GP adapter, is processed in order to learn specific temporal features. The first feature extraction step of our architecture processes each channel independently through a cascade of depth-wise separable 1D convolutions across time. In our case, the “depth” term refers to the EEG channel dimension. Employing separable convolutions allows the model to learn channel-specific patterns. In detail, the feature extraction module consists of a cascade of two blocks, each of which applies 1D convolutions with different dilation values (1, 2, 4, 8), concatenates the results and reduces the feature dimension through a pointwise convolution. Batch normalization is used in each layer, and the size of the convolution kernel is 3. Each block computes 32 feature maps; in our experiments, using a larger values showed signs of overfitting during training. After each block, the temporal dimension is halved.

D. Spatio-temporal self-attention

Once we have features for each channel of the EEG input, a spatio-temporal self-attention block is applied. This module evaluates the attention between all pairs of feature vectors, as computed by the feature extraction part, over both the spatial and temporal dimensions. Separately handling the temporal

attention is one of the key aspects of our approach: for instance, self-attention may find that the activity of channel i at time t_1 is not affected by channel j at the same time, but is instead affected by j at a different time t_2 , as is often the case for electrodes located over different scalp regions. A new set of features and the matrix of attention scores between all pairs of features vectors are returned, and these are used for further processing.

Self-attention is implemented by means of a non-local block [18]. Given an input tensor $\mathbf{f} \in \mathbb{R}^{F \times C \times T}$, where F is feature dimensionality, C is the number of channels and T is the (aggregated) temporal dimension resulting from the above feature extractor, three 1×1 convolutional operations (operating on each spatio-temporal location individually) are applied on $\mathbf{f}$ to compute three representations $\theta(\mathbf{f})$, $\phi(\mathbf{f})$ and $g(\mathbf{f})$. After appropriate reshaping (such that $\theta(\mathbf{f}) \in \mathbb{R}^{CT \times F}$ and $\phi(\mathbf{f}) \in \mathbb{R}^{F \times CT}$), pairwise affinity can be computed via matrix multiplication, and location-wise attention scores w.r.t. all other locations can be normalized by applying softmax to each row of the resulting product:

$$\mathbf{A} = \theta(\mathbf{f}) \cdot \phi(\mathbf{f}) \quad \mathbf{S} = \text{softmax}(\mathbf{A}) \quad (4)$$

Attention scores are used to weigh each row of the reshaped $g(\mathbf{f}) \in \mathbb{R}^{CT \times F}$ representation at each spatio-temporal location, by computing a linear combination between features at each location based on the corresponding row in $\mathbf{S}$. A final pointwise convolution and a residual sum with the original input provides the output features of the non-local layer.

Non-locality in this operation is due to the fact that all spatio-temporal locations in the input feature map $\mathbf{f}$ are considered in the operation. This allows us to capture long-range dependencies in space and time between EEG channels, not only by employing self-attended features provided as output by this block, but also by taking into account the affinity matrix $\mathbf{A}$, as described in the following.

E. Graph-based classification

Graph-based convolutions [23] are commonly employed to extract features from inputs whose topology cannot be adequately represented by regular grids. EEG signals can benefit from such models due to the non-uniform distribution of electrodes over the scalp. Furthermore, a graph can establish connections between EEG channels that are not spatially or temporally close, which allows us to model time-delayed processes between different brain areas. A graph convolutional layer is defined as

$$\mathbf{y} = f\left(\mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2} \mathbf{x} \mathbf{W}\right) \quad (5)$$

where f is the activation function (e.g., ReLU), $\mathbf{A}$ is the adjacency matrix, $\mathbf{D}$ is the diagonal node degree matrix of $\mathbf{A}$, $\mathbf{x}$ is the input feature tensor, and $\mathbf{W}$ refers to the learnable weights. Intuitively, this operation can be seen as a regular nonlinear layer in a neural network, whose input is a linear combination of features from all nodes, normalized and weighted by the adjacency matrix.

Our model uses two branches of graph convolutional layers, as shown in Fig. 1. Both branches are designed to extract features at long ranges between channels and times, in two different but complementary ways.

In the first branch (top of the figure), the inputs to the graph convolutional layer are attention-weighted spatio-temporal features provided by the non-local block. This network branch is meant to emphasize patterns that occur locally: while using features that represent collective information, we force a local connectivity to the graph by employing an adjacency matrix whose elements depend on the distance between the corresponding electrodes: $a_{ij} = \exp(-\|\mathbf{p}_i - \mathbf{p}_j\|)$. Then, for an input feature map of size $F \times C \times T$, we introduce C nodes in the graph: the input feature vector to each node corresponds to the FT -dimensional concatenation of temporal features for a certain channel. Due to the non-locality introduced by the self-attention layer, these features combine information from all EEG channels, regardless of distance.

In the second branch (bottom of the figure), node features are initialized using the temporal features computed through 1D separable depth-wise convolutions (thus, they are channel-specific). In this case, we design a graph with CT nodes, such that each node represents a single channel at a specific (aggregated) time step. Node features are initialized as the vectors along the first dimension of the $F \times C \times T$ tensor computed in the feature extraction stage. These channel-specific features are connected by employing the affinity matrix $\mathbf{A}$ from the non-local block as adjacency matrix, modified by adding 1 in the diagonal, in order to introduce self-loops.

In order to make the adjacency matrices sparse and reduce overfitting, we threshold both adjacency matrices to remove low-weight connections. [24] observed that sparse fMRI networks, comprising around 20% of all possible connections, typically maximize the efficiency of the network topology. Accordingly, we threshold each adjacency matrix to keep the strongest 20% of the entries.

IV. EXPERIMENTAL RESULTS

As a benchmark for evaluating the performance of the proposed method, we employ the SEED dataset [25], which targets emotion recognition. Each subject in this dataset was recorded while watching 72 2-minute video clips, labeled with the emotion they are expected to induce in the watcher (positive, negative, and neutral). A total of 15 subjects participated in the experiment. For each participant, 3 sessions were performed on different days, and each session contains 24 trials. In each trial, the participant watched one of the film clips, while recording their neural activity with a 62-channel ESI NeuroScan System. The dataset released by the authors includes both raw signals and differential entropy features.

We specifically used differential entropy (DE) features [26] typically adopted in EEG emotion recognition tasks [27]. Differential entropy for any random variable X is expressed as follows:

$$H(X) = - \int_{\mathcal{X}} f(x) \log(f(x)) dx \quad (6)$$

When X is a time-series with Gaussian-distributed values, i.e. $X \sim \mathcal{N}(\mu, \sigma^2)$, the expression simplifies to

$$H(X) = \frac{1}{2} \log(2\pi e \sigma^2) \quad (7)$$

Although EEG signals do not follow a fixed distribution, over small frequency intervals they are approximately Gaussian [26]. Hence, following common practice, we extracted DE features on sub-bands: δ (1–3 Hz), θ (4–7 Hz), α (8–13 Hz), β (14–30 Hz), and γ (31–40 Hz), estimating the standard deviation on segments of 125 ms (32 samples).

We treat the emotion recognition task at hand as a supervised classification problem, and train the model with mini-batch gradient descent using an Adam optimizer [28] with default hyperparameters (β_1 : 0.9, β_2 : 0.999), a learning rate 0.001 and batch size 32. L_2 regularization, with a weight decay of $5 \cdot 10^{-4}$. The number of learned channel locations in the GP adapter was set to 32.

The temporal length of the DE features with the SEED dataset was pre-computed by the dataset curators and is 180. We evaluated classification using a subject-independent analysis with a leave-one-subject-out cross-validation, as specified in [14]. Results, shown in Tab. I, show that our DGA model outperforms all other models using the SEED dataset. We have also conducted a number of ablation studies to assess the importance of each component in our architecture. Given the performance in Tab. I, the ablation studies were carried out using feature aggregation on multiple bands. We performed two sets of experiments. One ablates modules from the full DGA architecture, while the second assesses the impact of our graph-based approach compared to other graph methods. We define as *Baseline* the variant of our model that directly feeds DE features to a single topology-based graph convolution network. We then incrementally add to the baseline, in order, the GP adapter, the temporal convolution block, the self-attention mechanism to extract non-local features (but still with topology-based adjacency matrix; this is labeled *spatio-temporal attention*) and our full model, with the additional attention-based graph convolution branch. Our results, reported in Table II, indicate that our spatio-temporal mechanism substantially boosts accuracy.

We also compare our dynamic graph attention approach with a set of other graph models. For this analysis, from the architecture shown in Fig. 1, we keep the GP adapter and temporal convolution block, but replace the graph-based model with each of the following: 1) **GCNN**: a standard graph convolutional neural network [23] with a topology-based adjacency matrix, taking as input the extracted temporal features; 2) **GCNN+LSTM**: a standard graph CNN processing each feature at a time followed by an LSTM recurrent network, as in [29]; 3) **GCLSTM**: a graph CNN with recurrent nodes as in [30]; and 4) **GAT**, a graph model with an attention mechanism limited to nearby nodes, proposed in [15]. Results in Tab. III indicate that our DGA approach is more suitable to model brain connections than other existing GNN architectures. In addition, our attention mechanism significantly outperforms

the one in [15]: the reason may be that GAT only models spatial correlations, while the emotional processes we aim to decode are characterized by temporal correlations.

V. CONCLUSION

We here presented a deep architecture employing self-attention and graph convolutions to capture spatio-temporal patterns across EEG channels. Our approach is also thought to provide a natural mechanism for interpreting self-attention scores in terms of relative importance between channels.

The obtained results show that our approach yields satisfactory performance on the SEED benchmark, outperforming state-of-the-art models and other graph-based approaches. Moreover, model’s self-attention mechanism is used to link EEG channels in the classification graph and, through the analysis of these learned attention scores, the neural activity patterns associated with emotion recognition are identified. This indicates that the model’s attention scores provide insights into how different EEG channels are influencing each other, shedding light on the underlying spatio-temporal patterns of neural activation relevant to emotional processing. This interpretability analysis helps to understand the inner workings of the model and provides valuable information about the neural mechanisms underlying the emotion recognition process.

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TABLE I: Classification accuracy on the SEED dataset using DE features with subject-wise leave-one-out cross-validation. Best result in bold.

Model	Band features					
	δ	θ	α	β	γ	Feat. aggregation
EEGNet [7]	56.89 $\pm$ 08.09	52.59 $\pm$ 09.64	59.22 $\pm$ 07.19	61.93 $\pm$ 08.96	65.19 $\pm$ 10.07	66.81 $\pm$ 09.74
Syncnet [9]	53.48 $\pm$ 08.29	52.59 $\pm$ 08.26	57.33 $\pm$ 07.28	65.19 $\pm$ 10.81	67.55 $\pm$ 09.83	63.40 $\pm$ 13.38
ChannelNet [10]	52.89 $\pm$ 07.84	53.18 $\pm$ 08.62	54.36 $\pm$ 09.38	69.48 $\pm$ 06.89	66.67 $\pm$ 06.50	64.45 $\pm$ 06.23
DGA	60.89 $\pm$ 04.66	58.67 $\pm$ 08.26	61.48 $\pm$ 06.70	70.96 $\pm$ 05.54	72.59 $\pm$ 08.96	70.07 $\pm$ 09.86

TABLE II: Ablation study on the SEED dataset using aggregated features over all bands.

Model	Accuracy
Baseline	57.47
+ GP adapter	60.52
+ Temporal convolution	54.14
+ Spatio-temporal attention	67.91
+ Attention-based adjacency (DGA)	70.07

TABLE III: Accuracy with different graph CNN architectures on the on the SEED dataset using aggregated features over all bands.

Model	Accuracy
Graph model variants	
GCNN	56.87
GCNN + LSTM	61.24
GCLSTM	63.82
GAT	54.67
DGA (our model)	70.07

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